Term 2 Checklist

Topic	Learning Outcomes	RAG Yourself	
		1	2
Y7 Energy (***revision from last year***)	Name the different forms of energy store e.g. kinetic, thermal		
	Draw energy transfer diagrams for a candle, a catapult and a TV		
	Identify useful and wasted energy in an energy transfer		
	Calculate the percentage efficiency of an energy transfer		
	Describe the difference in terms of energy transfer of an efficient device and a less efficient device		
	State the meaning of "non-renewable"		
	State the tree types of fossil fuel and describe their uses		
	Explain how fossil fuels are used to generate electricity in a power station		
	Define what a renewable energy resource is		
	State examples of renewable energy resources e.g. solar, wind		
	Explain how solar, wind and tidal resources work in terms of energy transfer		
	Compare the advantages/disadvantages of renewables and fossil fuels		
Human Body	Draw labelled diagrams of plant and animal cells		
	Explain the importance of mitochondria in releasing energy		
	Define diffusion and give examples of diffusion in the human body		
	Describe how to calculate rate of diffusion in cm/s		
	State the 3 main nutrients and describe their uses in the body		
	Explain how temperature and concentration changes affect rate of diffusion		
	Explain why starch is an important storage molecule		
	Describe how to perform an iodine test for starch and interpret the results		
	Describe how to perform a Benedict's test for sugar and interpret the results		
	Name the major digestive organs and describe their functions		
	Describe the order food travels through the digestive system		
	Describe where chemical and mechanical digestion take place		
	Explain the importance of digesting the food we eat		
	Describe how enzymes are affected by changes in temperature		
	Suggest why digestive enzymes only work in certain areas of the body		
	Explain the importance of villi in the digestive system		
	Describe the function of the circulatory system		
	Describe the structure and function of the heart		
	Explain the different thicknesses of the left and right walls of the heart		
	Explain why a high fat diet can lead to an increase in blood pressure		
	Label the main parts of the lungs on a diagram		
	Describe and explain the adaptations shown by the alveoli		
	Explain the effect of smoking on lung function		
	Explain how breathing works in terms of changes in pressure in the chest		
	Describe the effects of the different components of a cigarette on the body		
	Describe the effects of stimulants on the body		
Heating and Cooling	Recall the three states of matter and the four changes of state		
	State the boiling and freezing point of water in degrees celsius		
	Describe the how the particles are arranged differently in solids, liquids and gases		
	Explain the properties of solids, liquids and gases in terms of the arrangement of particles e.g. why can gasses be compressed?		
	Describe what is meant by expansion and contraction		
	Describe what happens to the particles in a substance when it is heated		
	Describe how thermal energy transfers between hotter and colder objects		
	Give the standard units of temperature and energy		

	Compare temperature to thermal energy in terms of kinetic energy of particles		
	Explain how colder objects can have more thermal energy than hotter objects		
	Describe how thermal energy transfer occurs through solids through conduction		
	Define the terms insulator and conductor with examples of each		
	Explain why metals are good conductors of energy		
	Describe how thermal energy transfers through liquids and gases by convection		
	Explain why the colour of a house is important when considering thermal insulation		
	Describe three methods used to reduce heat transfer in a house		
	Explain insulation methods in terms of how they reduce conduction, convection or thermal radiation from a house		
	Define the principle of conservation of energy		
	Describe how temperature of a substance changes when a substance is heated past its melting point		
	Explain the shape of a cooling or heating curve		
	Explain why evaporation leads to cooling in terms of the energy of particles lost		
	Describe what is meant by "rate of change" and how to calculate the rate of change in temperature		
	Explain why a hot substance cools more quickly if broken into smaller pieces		
	State the equation used to calculate pressure		
	Describe how gases create pressure in a sealed container in terms of particles		
	Explain why the impact of changing temperature on gas pressure		